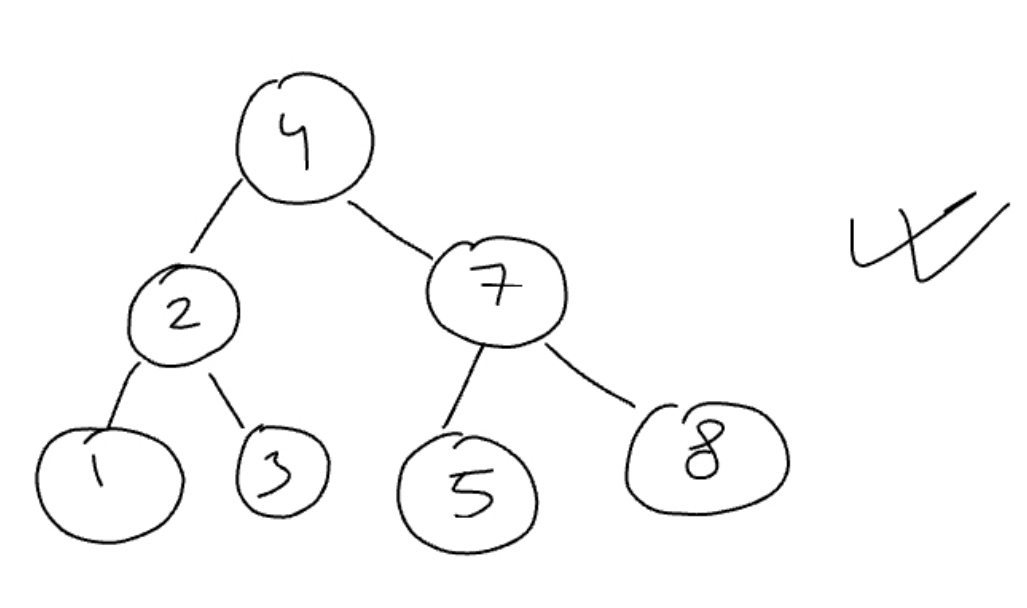
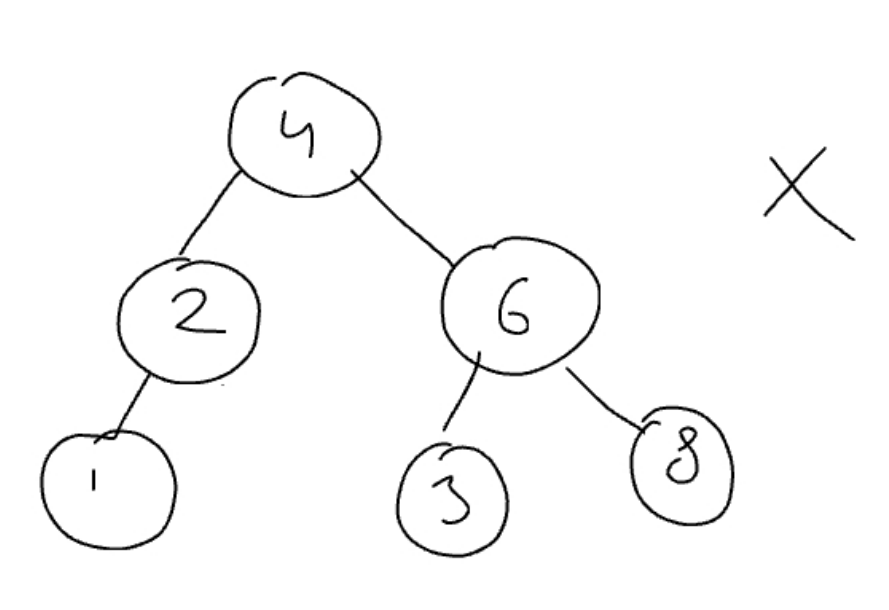
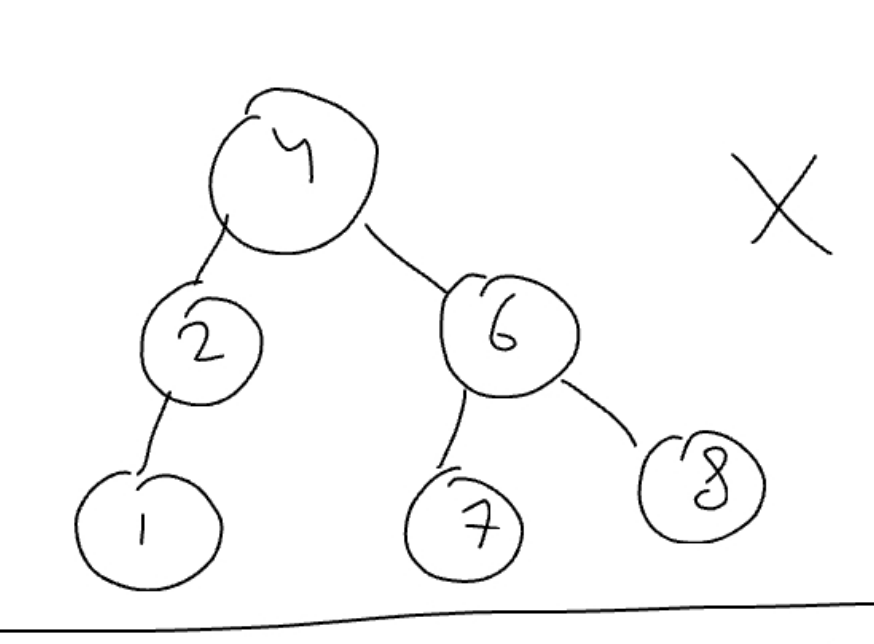
# Binary Search Tree (BST)







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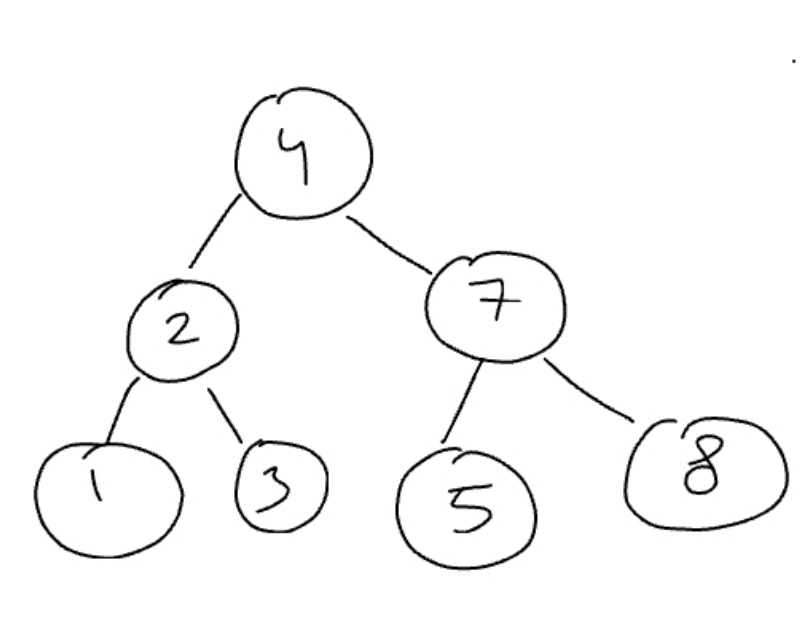
## Inorder



[1, 2, 3, 4, 5, 7, 8].

# Inorder traversal of a BST is always sorted.

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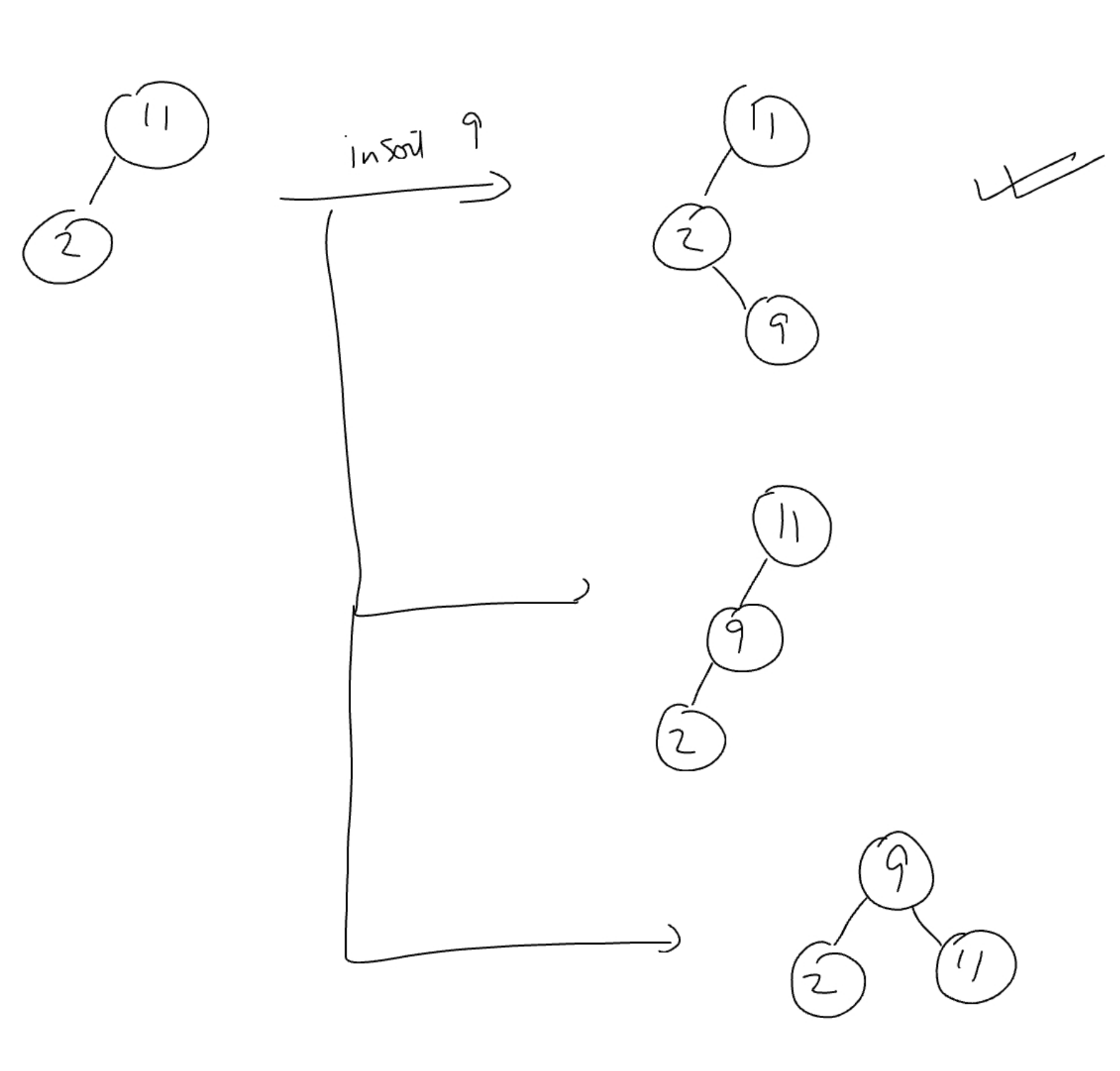
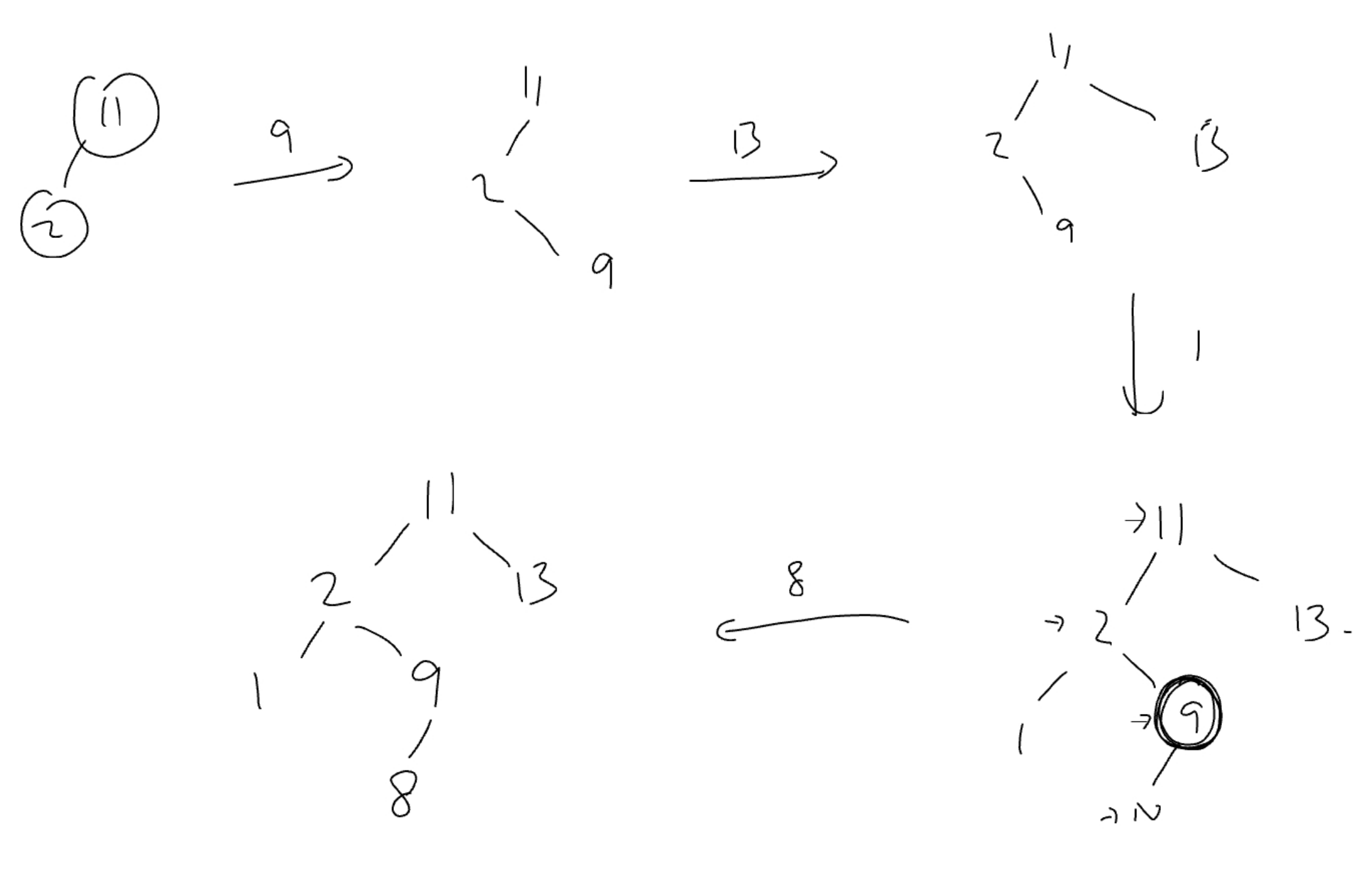
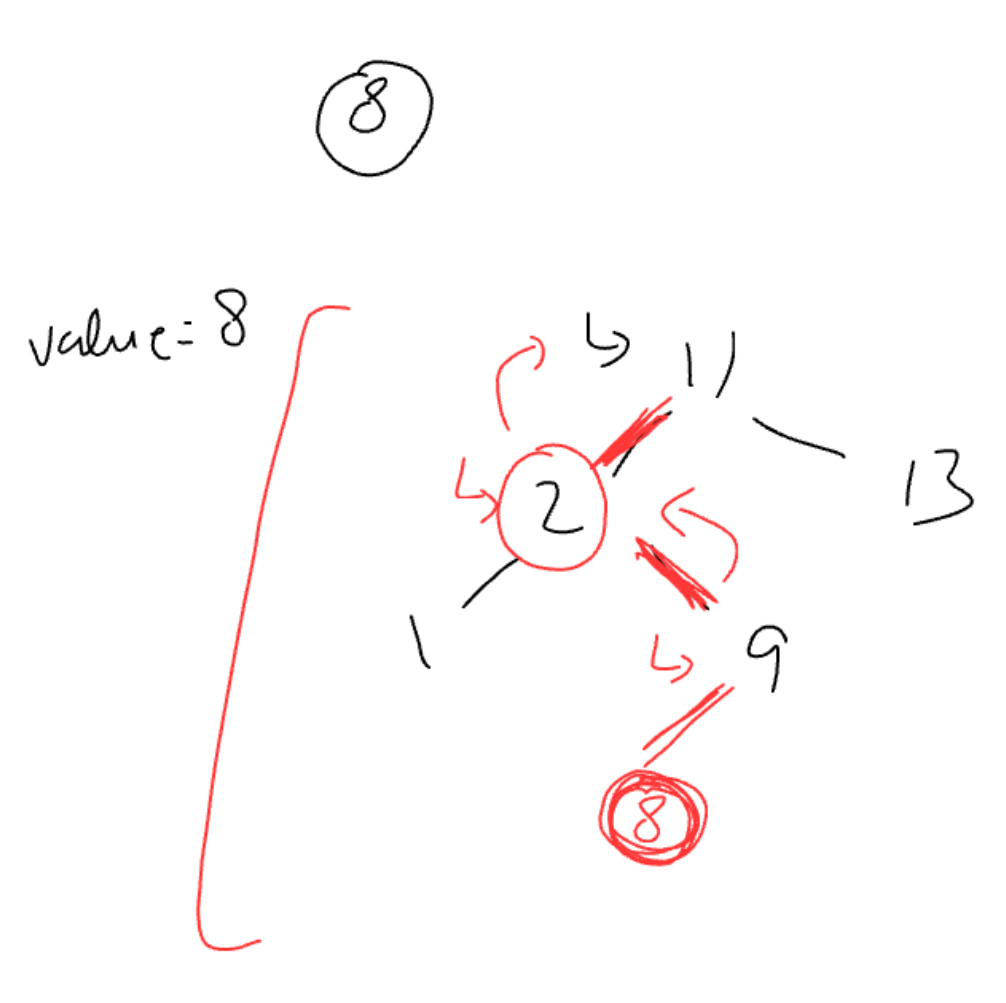
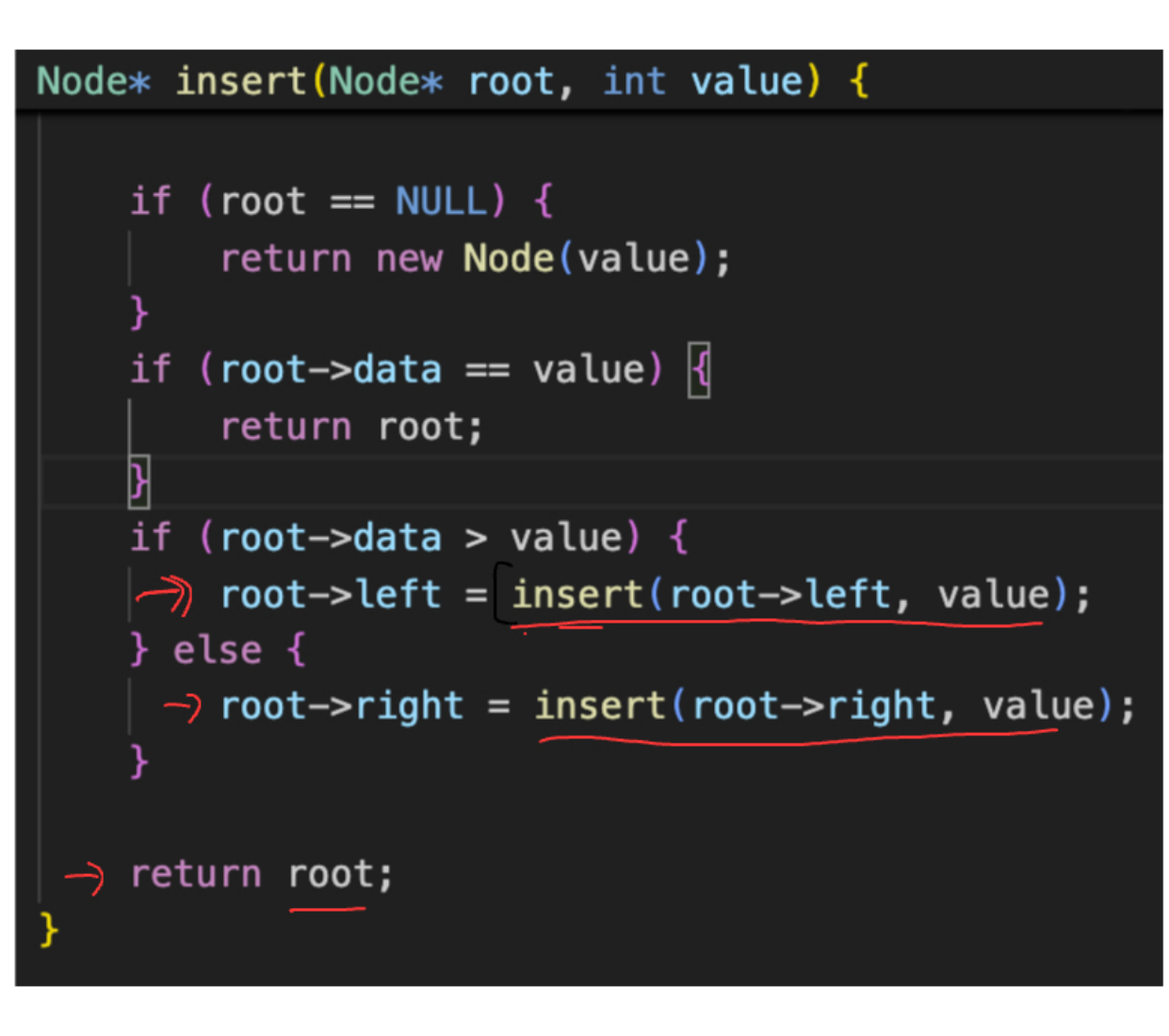
## Insert in a BST.


11
2
insert 9
11
2
9
✓✓
11
9
2
9
2
11


11
2
9
11
2
9
13
11
2
13
9
1
→11
→2
13.
1
→9
→N
8
11
2
13
1
9
8

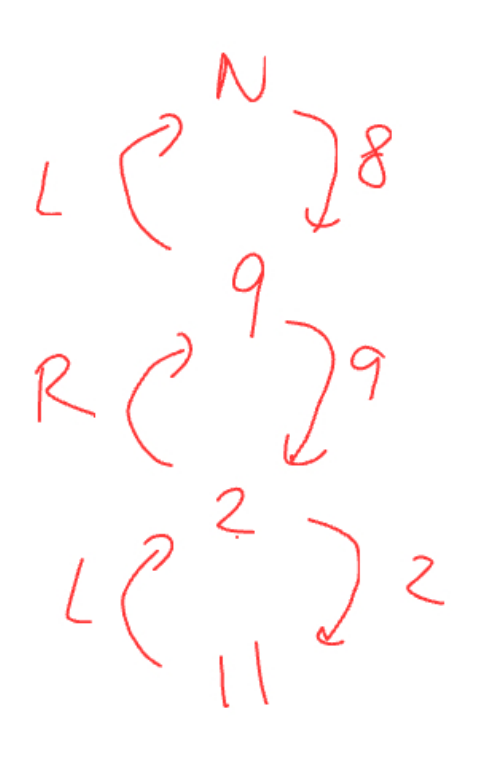

8
value: 8
11
2
13
1
9
8



```
Node* insert(Node* root, int value) {

    if (root == NULL) {
        return new Node(value);
    }
    if (root->data == value) {
        return root;
    }
    if (root->data > value) {
    →   root->left = insert(root->left, value);
    } else {
     →  root->right = insert(root->right, value);
    }

  → return root;
}
```


N
L
8
9
R
9
2
L
2
11

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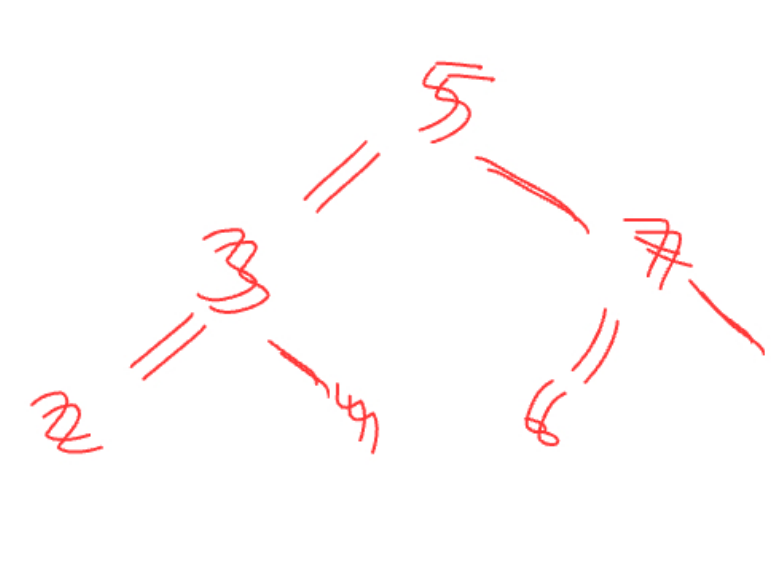
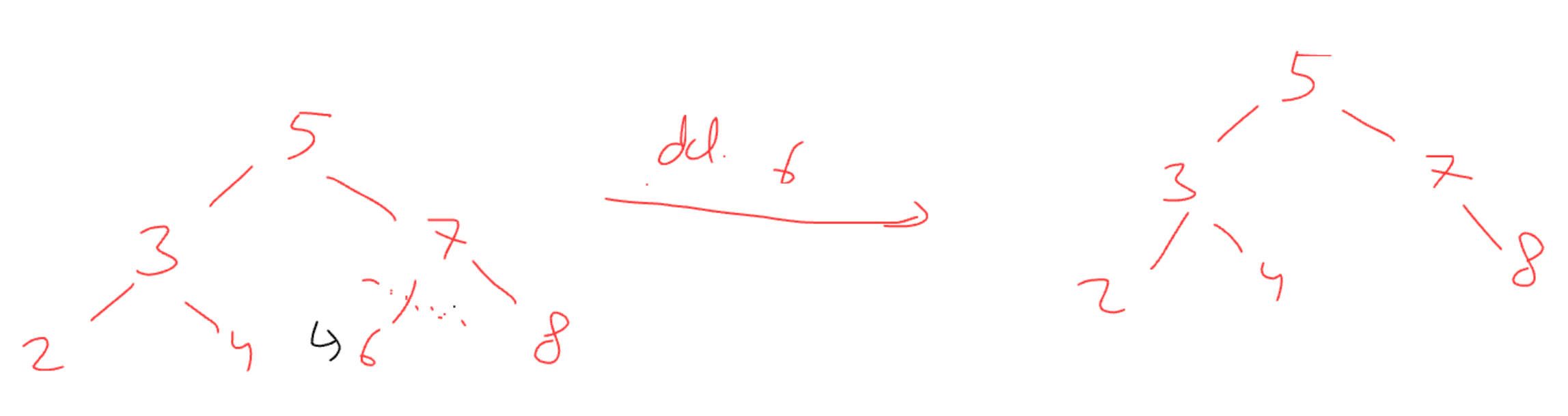
## Delete.

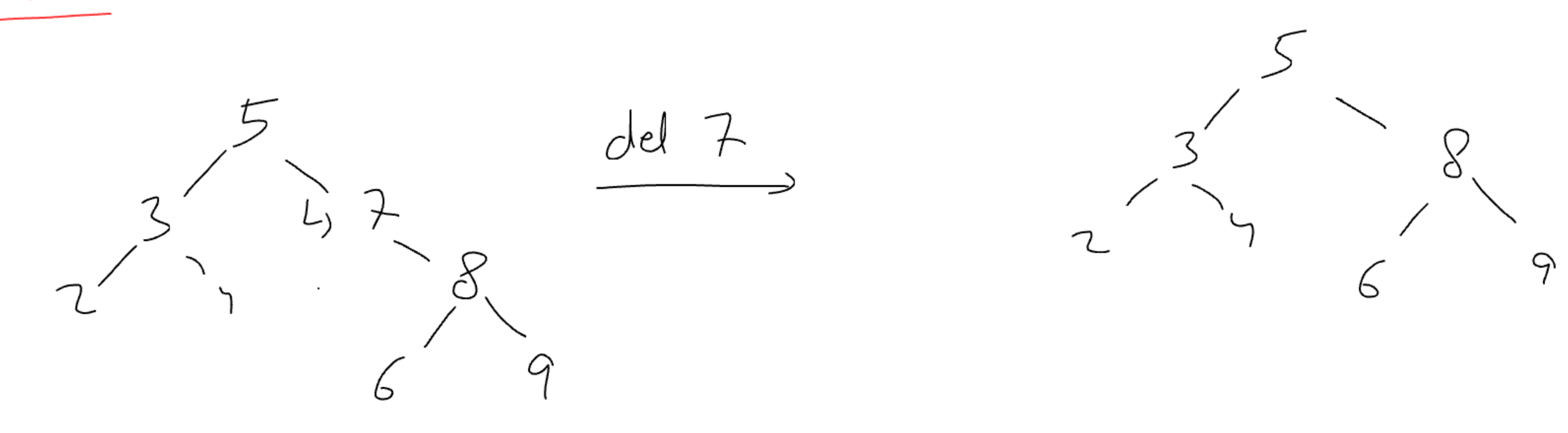


Case-1. Node to be deleted has 0 child nodes


5
3
7
2
4
6
8
del 6
5
3
7
2
4
8

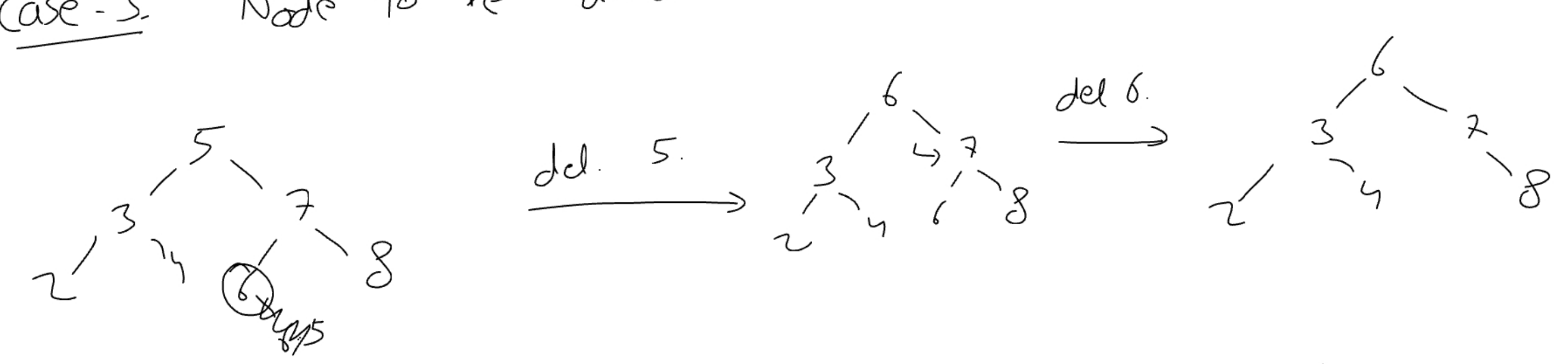
Soln: Replace the node with NULL.

Case-2. Node to be deleted has 1 child node.

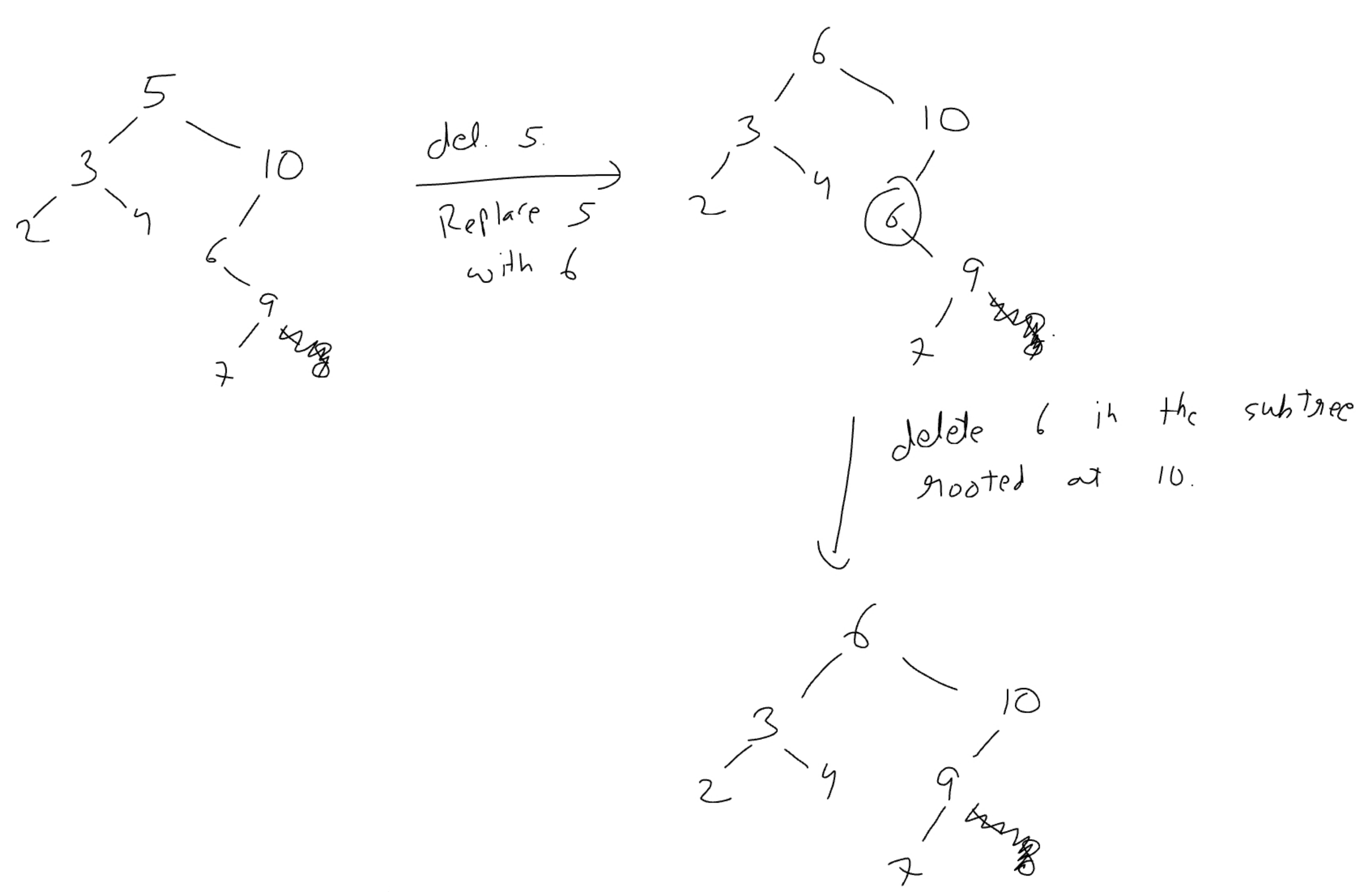

5
3
7
2
4
8
6
9
del 7
5
3
8
2
4
6
9

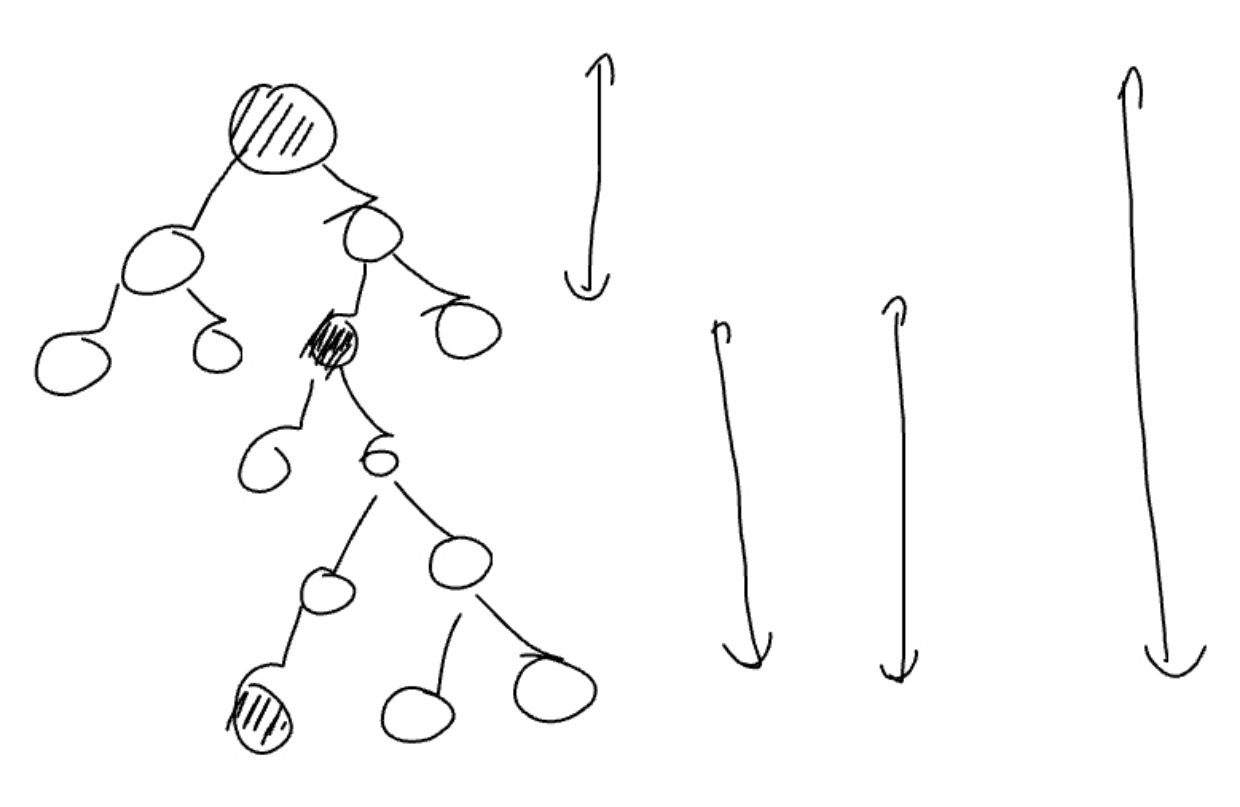
Soln. Replace the node with its non-NULL child node.

Case-3. Node to be deleted has 2 child nodes


5
3
7
2
4
6
8
del 5.
6
3
7
2
4
8
del 6.
6
3
7
2
4
8

Soln. Replace the node with its inorder successor / predecessor.
& then delete the inorder successor / pred.


5
3
10
2
4
6
9
7
8
del 5
Replace 5 with 6
6
3
10
2
4
6
9
7
8
delete 6 in the subtree rooted at 10.
6
3
10
2
4
9
7
8



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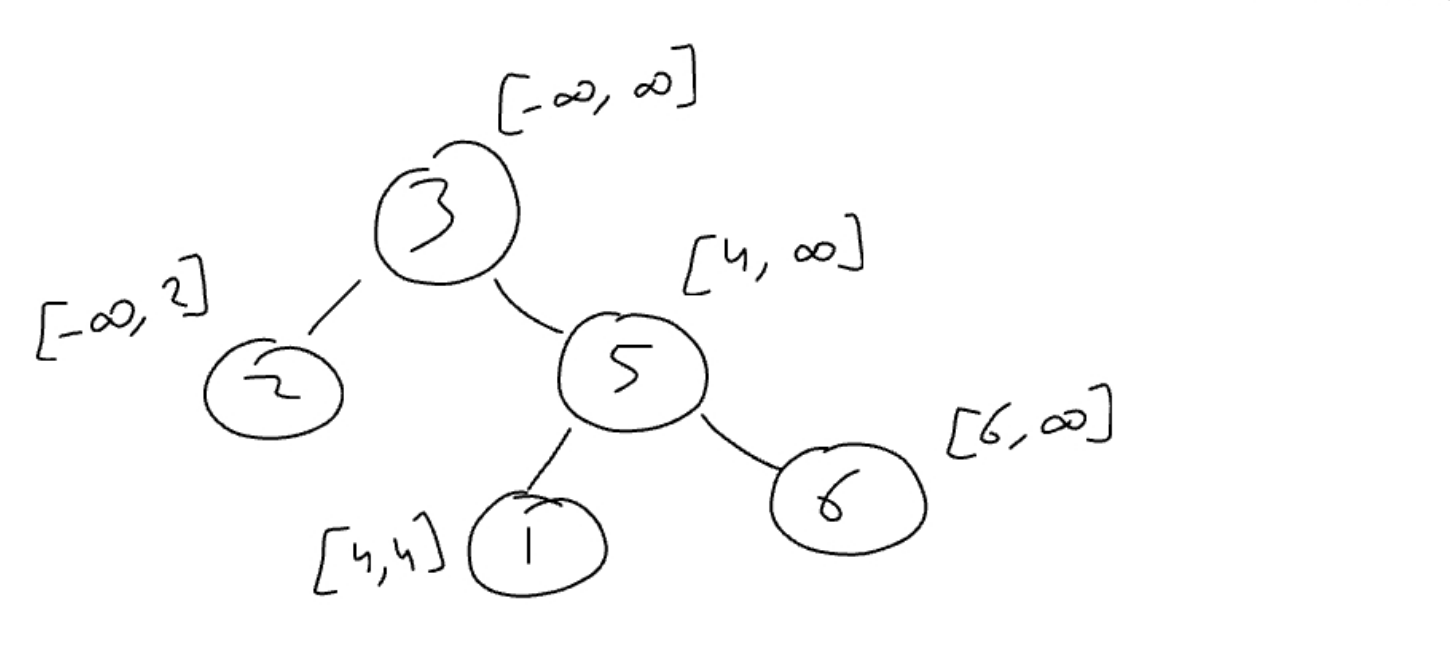
[-∞, ∞]
3
[-∞, 3]
2
[4, ∞]
5
[4, 4]
1
[6, ∞]
6

BF. For every node, find the largest in the LT
& smallest in the RT.

largest from LT < Root < smallest from RT.

TC = O(n²).
AS = O(H).

Inorder_Based. TC = O(n)
AS = O(n) → O(H).

Range_Based. TC = O(n)
AS = O(H) ✓✓.

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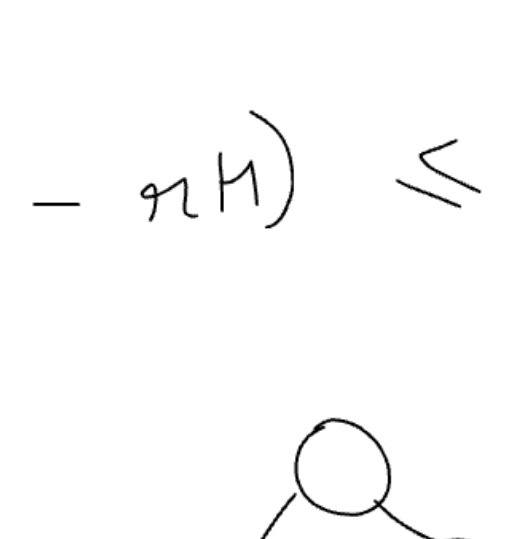
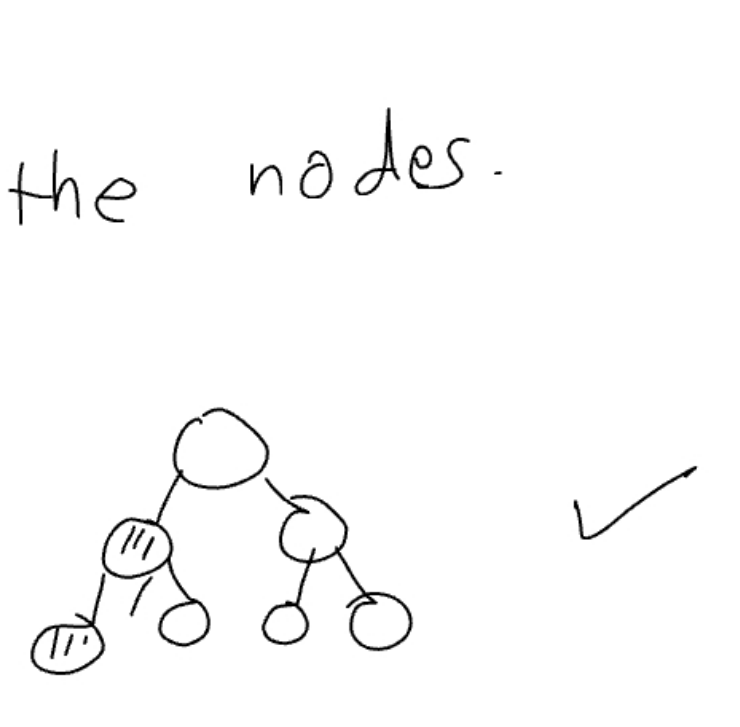
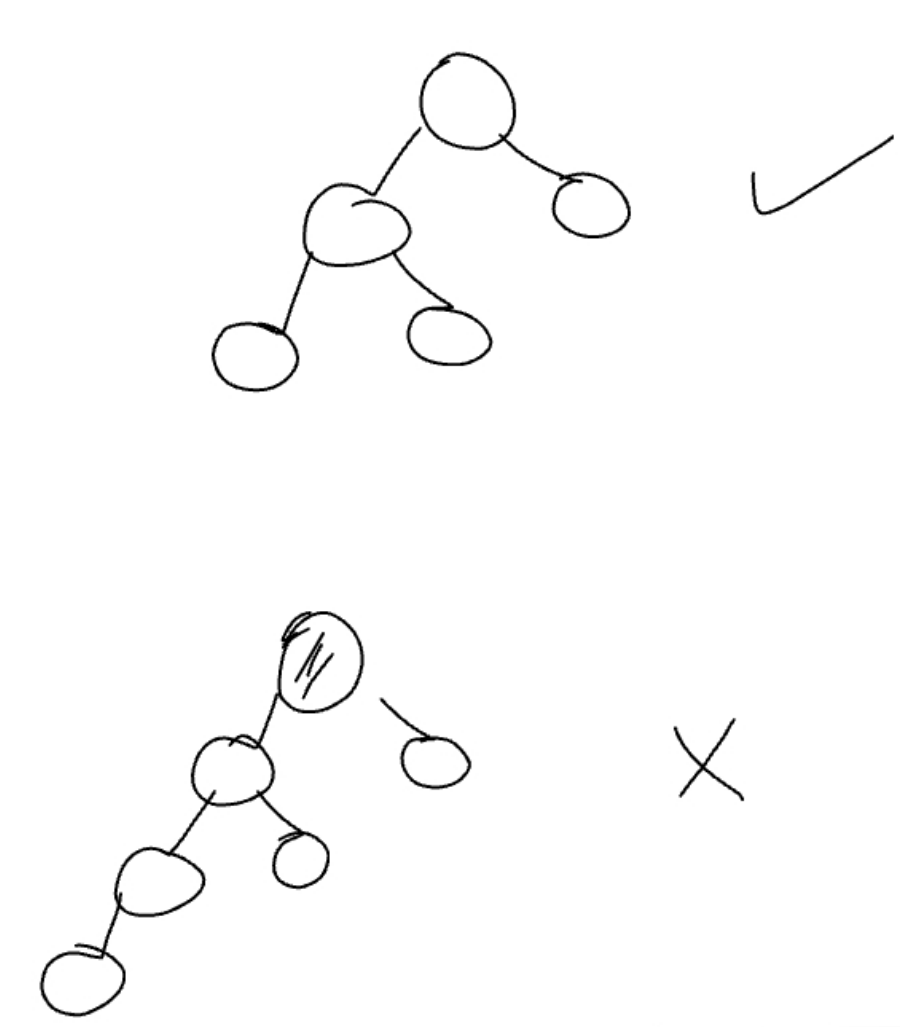
## Height_Balanced_BST.

abs(lH − rH) ≤ 1 for all the nodes.







For Height-Balanced Trees, Height is $O(\log n)$.

Ordered maps/sets use height-Balanced BSTs internally.

| | Unordered | Ordered |
|---|---|---|
| Insert | $O(1)$ | $O(\log n)$ |
| Search | $O(1)$ | $O(\log n)$ |
| Delete | $O(1)$ | $O(\log n)$ |

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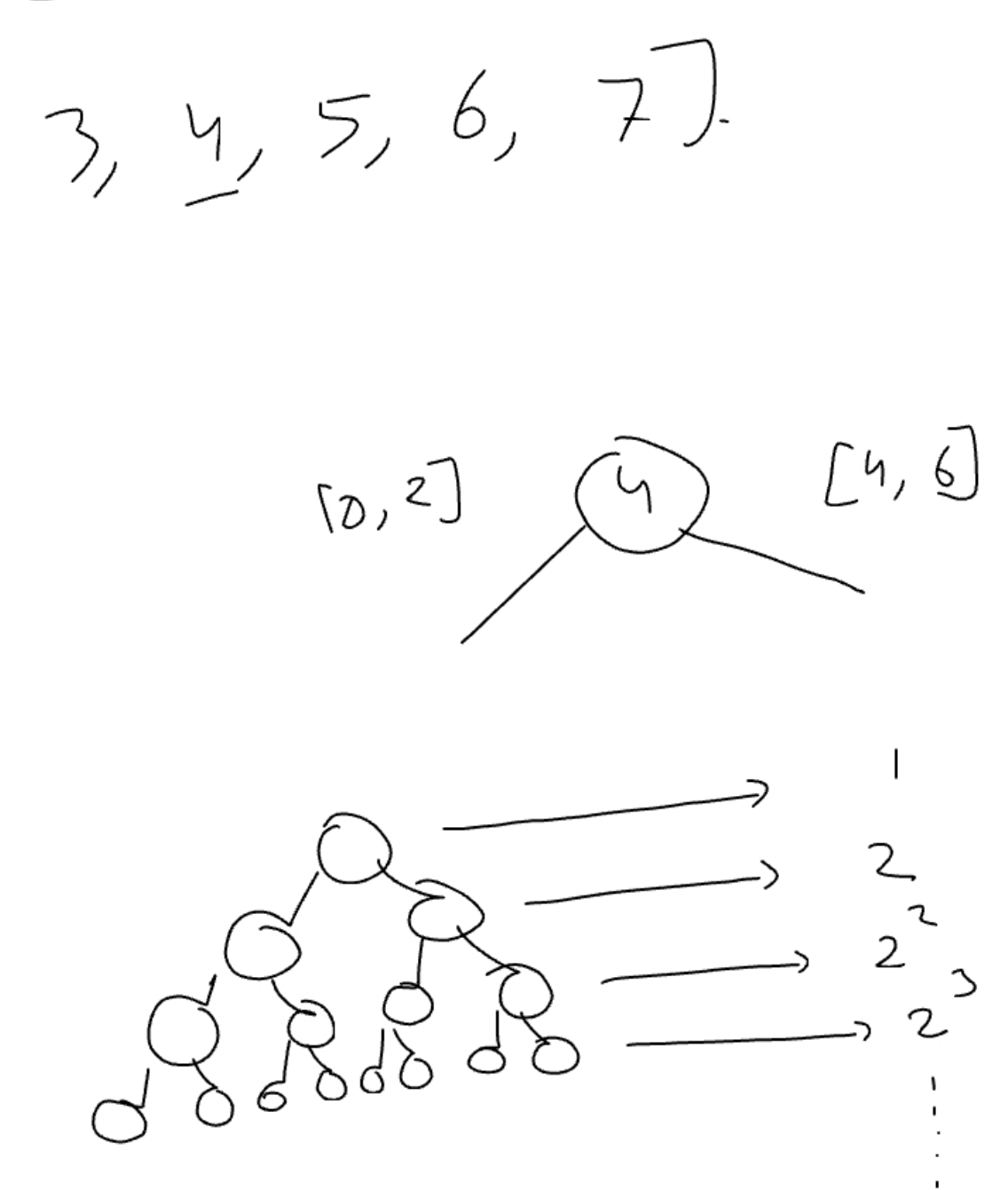
0 1 2 3 4 5 6
[1, 2, 3, 4, 5, 6, 7].


[0, 2]
4
[4, 6]
1
2
2²
2³
⋮

$2^0 + 2^1 + 2^2 + 2^3 + \dots + 2^{H-1} = n$

$1 \cdot \frac{(2^H - 1)}{2 - 1} = n$

$2^H \approx n$

$\boxed{H = \log_2 n}$ for Height-Balanced BSTs